Q1: Is there a fault in the program? If so, locate it and explain.

Yes, there is a fault in the program located in the for loop initialization statement: for (int i = 1; i < attendance_records.size(); ++i). C++ vectors use 0-based indexing so a valid attndance record for 10 lectures contains indices ranging from 0 to 9. By starting the loop at i = 1, the program actually completely skips the first lecture at index 0 so any absence on the first lecture is never counted, which causes a logic issue.

Q2: Define a test case that does not execute the fault.

It's impossible to define a test case that doesn't execute the fault. The fault is the loop initialization statement int i = 1, which is the first part of the for statement and runs every time fail_lecture is called, before the loop condition is even checked. It does not depend on the contents of attendance_records, so every possible input executes it. Even if the loop never ran, the initialization would still execute. So, every test case executes the fault, and whether it leads to an error state or a failure depends only on the input data and not on whether the fault is reached.

Q3: Define a test case that executes the fault but does not result in an error state.

A test case that executes the fault but doen't result in an error state would be where the attendance record is {1, 1, 1, 1, 1, 1, 1, 1, 1, 1} with 0 total absences, an expected absence count of 0, an actual absense count of 0, an expected output of false, and an actual output of false. Even though the loop still starts at i = 1 and executes the fault, the student attended every lecture including index 0, so skipping index 0 does not alter the count, keeping the internal state correct and causing the no error state.

Q4: Define a test case that results in an error state but not a failure.

A test case that results in an error state but not a failure record is {0, 1, 1, 1, 1, 1, 1, 1, 1, 1} with 1 total absence located at index 0, an expected absence count of 1, anabsence count of 0, an expcted output of false, and an actual output of false. The fault is executed and skips the absence at index 0, causing the internal "absent_count" variable to become 0 instead of 1, which puts the program into an incorrect error state. However, since the threshold for failing is 3 or more absences, both the expected outcome (1 is not greater than or equal to 3) and actual outcome (0 is not greater than or equal to 3) evaluate to false, meaning the external output matches and no failure occurs.

Q5: Define a test case that results in failure.

One test case would be the attendance record {0, 0, 0, 1, 1, 1, 1, 1, 1, 1} with 3 total absences at indices 0, 1, and 2, an expected absence count of 3, an actual absence count of 2, an expected output of true, and an actual output of false. The fault is executed and skips index 0, resulting in an incorrect internal absent_count of 2 instead of 3. This internal error spreads to the return statement where 2 >= 3 evaluates to false instead of true, causing the external behavior to deviate and resulting in a failure.